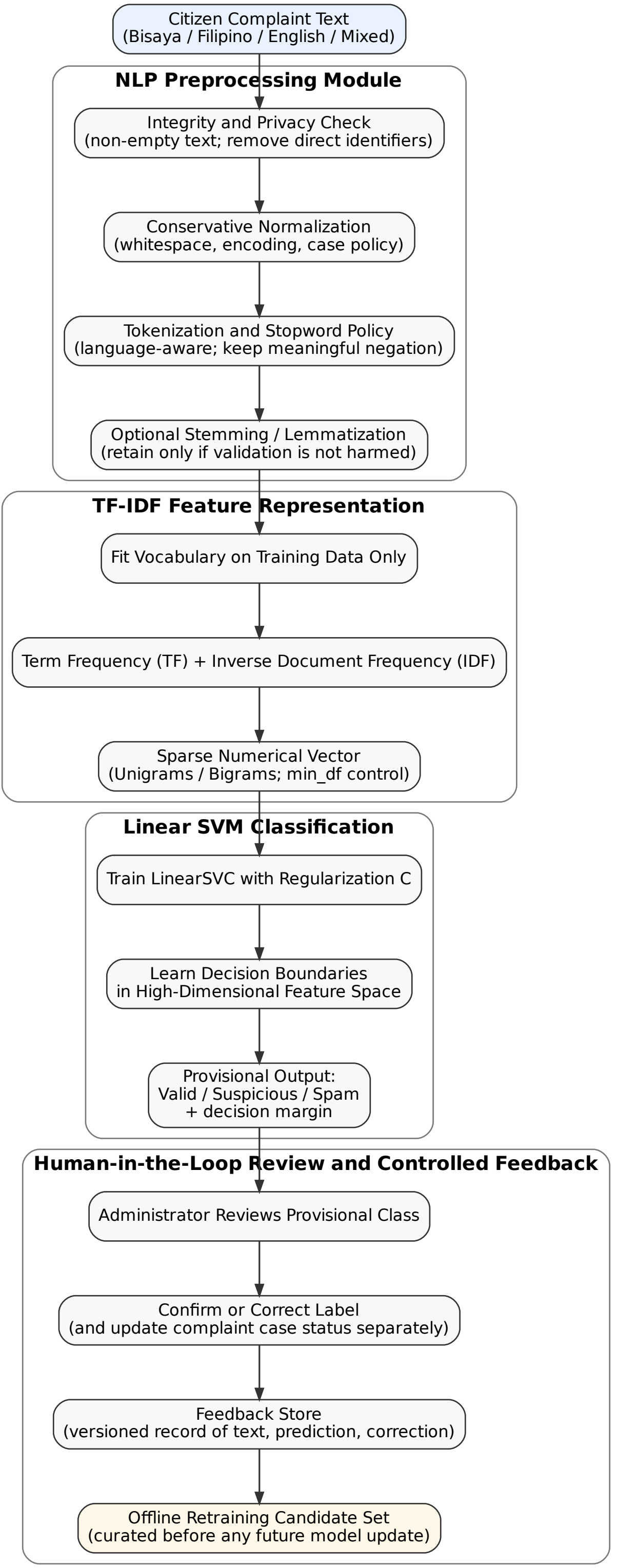
Citizen Complaint Text
(Bisaya / Filipino / English / Mixed)
NLP Preprocessing Module
Integrity and Privacy Check
(non-empty text; remove direct identifiers)
Conservative Normalization
(whitespace, encoding, case policy)
Tokenization and Stopword Policy
(language-aware; keep meaningful negation)
Optional Stemming / Lemmatization
(retain only if validation is not harmed)
TF-IDF Feature Representation
Fit Vocabulary on Training Data Only
Term Frequency (TF) + Inverse Document Frequency (IDF)
Sparse Numerical Vector
(Unigrams / Bigrams; min_df control)
Linear SVM Classification
Train LinearSVC with Regularization C
Learn Decision Boundaries
in High-Dimensional Feature Space
Provisional Output:
Valid / Suspicious / Spam
+ decision margin
Human-in-the-Loop Review and Controlled Feedback
Administrator Reviews Provisional Class
Confirm or Correct Label
(and update complaint case status separately)
Feedback Store
(versioned record of text, prediction, correction)
Offline Retraining Candidate Set
(curated before any future model update)